

Noah Hess

Mechanical Engineering Student - Available for co-op positions starting May 2027

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EDUCATION

University of British Columbia

Expected: May 2028

Bachelor of Applied Science (Year 4 Standing) – Mechanical Engineering

- CGPA: 84% (Calculus: 98%, Computation: 96%, Engineering Design: 91%, Instrumentation: 91%)

WORK EXPERIENCE

Metro Vancouver – Project Research Assistant Co-op (Vancouver, BC)

September 2025 – April 2026

- Automated the ultraviolet radiation data verification workflow using a centralized Excel tool with dynamic references to cross-validate graphical and plant data and programmatically flag errant sensor readings and data transfer errors.
- Optimized predictive simulation accuracy for an alpine lake discharge model by benchmarking established governing equations against historical precipitation and operational data to tune loss coefficients.
- Authored a unified reference memo consolidating reservoir lake elevation parameters by auditing engineering drawings to extract missing information and verify historical benchmarks, ensuring cross-department consistency.
- Coordinated regional water supply maintenance operations by identifying and gathering supporting documentation from requestors and presenting updates and requests to the Utility Systems Control Superintendent to facilitate approvals.
- Developed cross-functional maintenance approval flowcharts that standardize review timelines and minimize administrative communication loops while clarifying approval requirements for project managers.

Genist Systems – Mechanical Design Engineer Intern (Vancouver, BC)

May 2025 – August 2025

- Spearheaded the initial mechanical design and architectural layout for a high-speed VTOL emergency response drone using Fusion 360, ensuring alignment with aerodynamic, structural, manufacturability and assembly requirements.
- Evaluated full-drone aerodynamic efficiency and validated internal structural components under critical loading conditions by performing CFD and structural FEA using SimScale and Ansys to guide and validate design decisions.
- Collaborated asynchronously with electrical, software, and manufacturing teams to integrate subsystem requirements, document design decisions, and manage version control of mechanical parts and configurations.

ENGINEERING PROJECTS

Excel Financial Tracker

October 2025 – November 2025

- Built an automated financial tracking workbook and summary dashboard utilizing pivot tables and consistent sheet-naming conventions to consolidate monthly, cumulative, and year-to-date metrics while visualizing spending patterns.
- Developed a portfolio management tool with variance-driven rebalancing logic to optimize ETF capital allocation based on real-time valuations and targets, while integrating visual performance graphs and tracking TFSA contribution room.

VTOL UAV Aileron Wing Design

April 2025 – May 2025

- Designed and modeled VTOL UAV wing featuring a Frise aileron and servo-actuated mechanism in Fusion 360 using a NACA 4412 airfoil based on performance requirements for speed, altitude, and payload.
- Performed CFD simulations in SimScale to evaluate lift, pressure distribution, and control-surface effects at various aileron deflections, verifying aerodynamic performance under specified operating conditions.

Alternate Power Case Study

September 2024 – December 2024

- Led stakeholder research and consultations with commercial lawn-care providers to identify user needs, validate engineering requirements, and define target specifications for non-ICE commercial lawn-mowing solutions.
- Developed quantitative satisfaction curves based on stakeholder input and external research, integrating them into a custom scoring calculator and value equation to support evidence-based evaluation of alternative power sources.
- Produced Gantt charts, risk registers, an NPV analysis, and an expression of interest for weekly progress reviews.

Firefighting Remotely-Operated Rough Terrain Vehicle

January 2024 – April 2024

- Designed a remote-controlled vehicle with a servo-actuated transmission and steering mechanism performing under simulated motor failure using PLA/metal hybrid construction for improved sustainability and performance.
- Engineered a compressed-air water cannon system using pressurized air/water vessels, servo-controlled valves, and a two-axis targeting mechanism capable of extinguishing elevated fires.

TECHNICAL SKILLS

Software: SolidWorks, Fusion 360, ANSYS, Python, C, C#, MATLAB, Office Suite, Google Workspace

Machining: Mill, Lathe, Power Tools